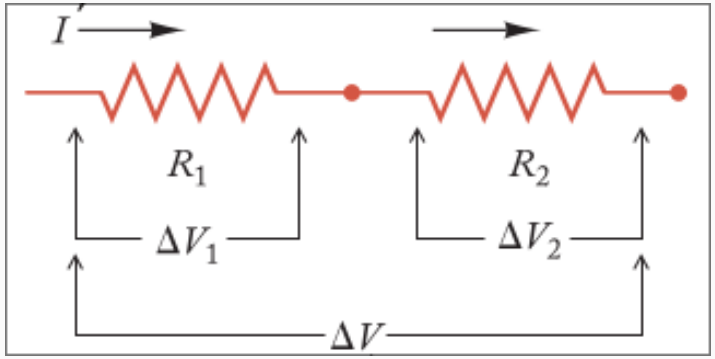
CHP-8 Circuits

General Physics 2 2026-10-26

Resistances in combination

Resistances in combination: series

Using Ohm's law $V = IR$, we can calculate the resultant resistance of a combination. Let us start with series.

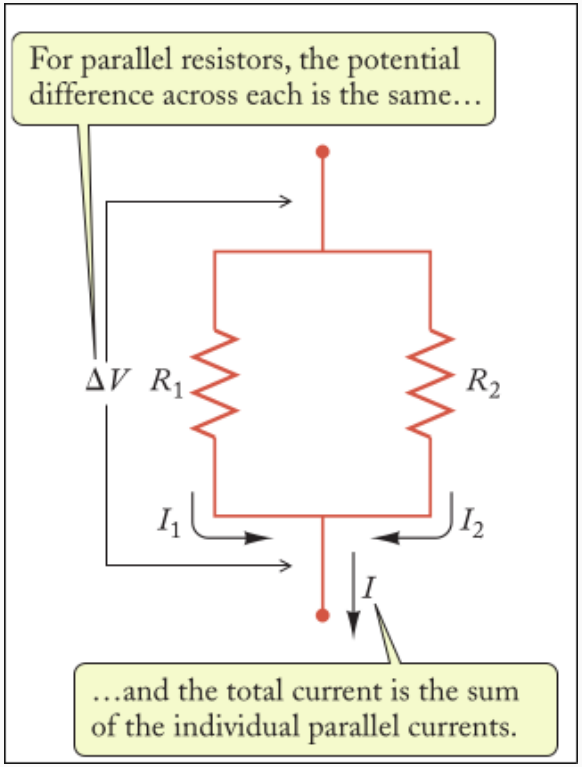


The potential drops across R_1 and R_2 are ΔV_1 and ΔV_2. The total drop from *a* to *b* is

$$\Delta V = \Delta V_1 + \Delta V_2 = IR_1 + IR_2 = I(R_1 + R_2) = IR$$

$$R = R_1 + R_2 + \ldots$$

Resistances in combination: parallel



The potential drop is the same across R_1 and R_2.

$$I_1 = \frac{\Delta V}{R_1}, \quad I_2 = \frac{\Delta V}{R_2}$$

By conservation of charge,

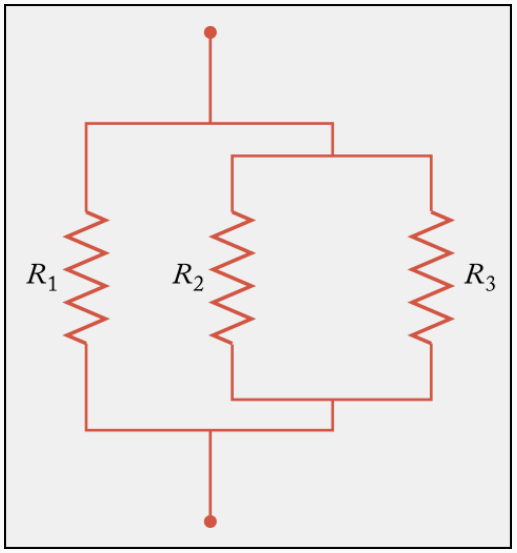
$$I = I_1 + I_2$$

$$\frac{\Delta V}{R} = \frac{\Delta V}{R_1} + \frac{\Delta V}{R_2}$$

$$\boxed{\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}}$$

Example: three resistors

Three resistors $R_1 = 4\,\Omega$, $R_2 = 6\,\Omega$, and $R_3 = 8\,\Omega$ are connected as shown.



(a) Find the net resistance of the combination.

(b) Find the total current if 12 V is applied.

(c) Find the potential drop across each resistor.

Example: net resistance and currents

First combine R_2 and R_3:

$$R_{23} = \left(\frac{1}{R_2} + \frac{1}{R_3}\right)^{-1} = \frac{R_2 R_3}{R_2 + R_3}$$

Then

$$\frac{1}{R} = \frac{1}{R_1} + \frac{R_2 + R_3}{R_2 R_3}$$

$$R = \frac{R_1 R_2 R_3}{R_2 R_3 + R_1(R_2 + R_3)} \approx 1.85\ \Omega$$

Example: total and branch currents

With 12 V applied across the combination,

$$I = \frac{12\ \text{V}}{1.85\ \Omega} \approx 6.5\ \text{A}$$

The potential drop is the same for all: $\Delta V = 12.0\ \text{V}$.

$$I_1 = 3\ \text{A}, \quad I_2 = 2\ \text{A}, \quad I_3 = 1.5\ \text{A}$$

DC circuits and emf

DC circuits

If we want to keep a current flowing in a wire, we must connect its ends to a **pump of electricity**, such as a battery or generator, that can continuously supply electric charges to one end and remove them from the other.

Such a steady, time-independent current is called a **direct current**, DC.

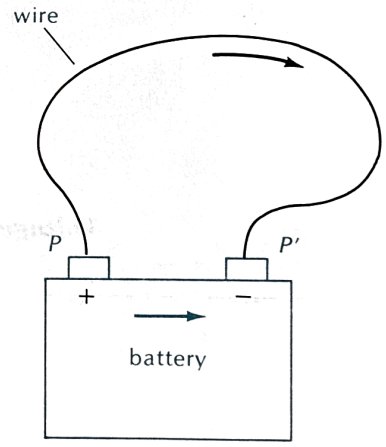


Fig. 29.1 A wire connected between the terminals of a battery.

A hypothetical positive charge is pushed by the electric field created by the battery. On average, its kinetic energy does not change because energy gains are dissipated by friction.

Electromotive force

Thus only potential energy changes. The potential steadily decreases, and the charge reaches P' at a lower potential. The battery must then pump charge to the high-potential terminal.

To measure the strength of the source of electric potential energy, we introduce **electromotive force**, or **emf**.

> Emf is the energy delivered by the source per coulomb of positive charge as the charge passes through the source from the low-potential terminal to the high-potential terminal.

Since emf is energy per coulomb, its unit is the volt.

Emf and the potential drop

For a steady, time-independent current, the energy supplied per unit charge by the emf ε and the potential drop have the same magnitude:

$$\varepsilon + \Delta V = 0$$

Thus emf is usually called the voltage of the source.

Sources of emf: batteries

Batteries convert chemical energy into electric energy. An old type is the lead-acid battery, still used in cars.

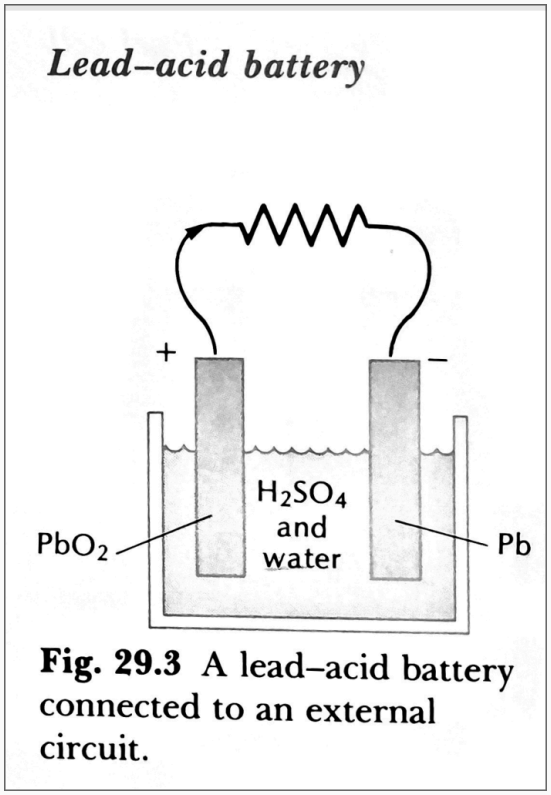


Fig. 29.3 A lead–acid battery connected to an external circuit.

At the negative electrode:

$$Pb + SO_4^{2-} \rightarrow PbSO_4 + 2e^-$$

At the positive electrode:

$$PbO_2 + SO_4^{2-} + 4H^+ + 2e^- \rightarrow PbSO_4 + 2H_2O$$

These reactions deplete sulfuric acid and deposit lead sulfate on the electrodes.

Recharging and other sources

If current passes in the opposite direction, we charge the battery.

The number of charges does not change; we change the concentrations of the chemicals.

Other examples of emf sources are dry cells, solar cells, and generators.

Single-loop circuits

Single-loop circuits

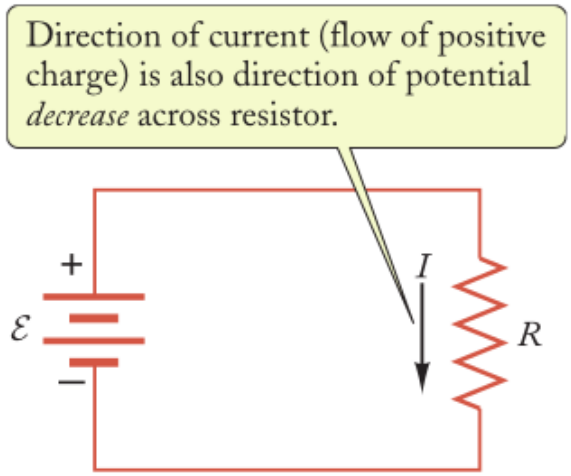


FIGURE 28.9 A simple circuit with a source of emf and a resistor.

According to Ohm's law, the potential drop across R is

$$\Delta V = -IR$$

Since $\varepsilon + \Delta V = 0$,

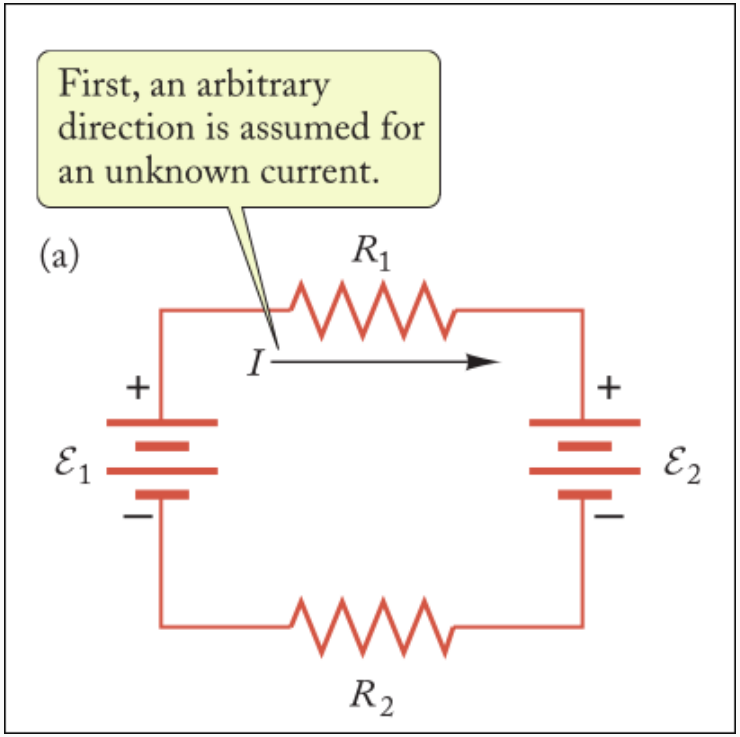
$$\varepsilon - IR = 0$$

This is an instance of Kirchhoff's loop rule:

Around any closed loop in a circuit, the sum of all emfs and potential changes must equal zero.

Signs and an assumed current direction

In the sum, the emf is positive when we pass through the source in its forward direction, from its negative to its positive terminal, and negative in the reverse direction.



To apply the rule, we must decide a direction for the current.

Assume clockwise current. If that assumption is wrong, the calculated current will be negative.

Example: opposing batteries

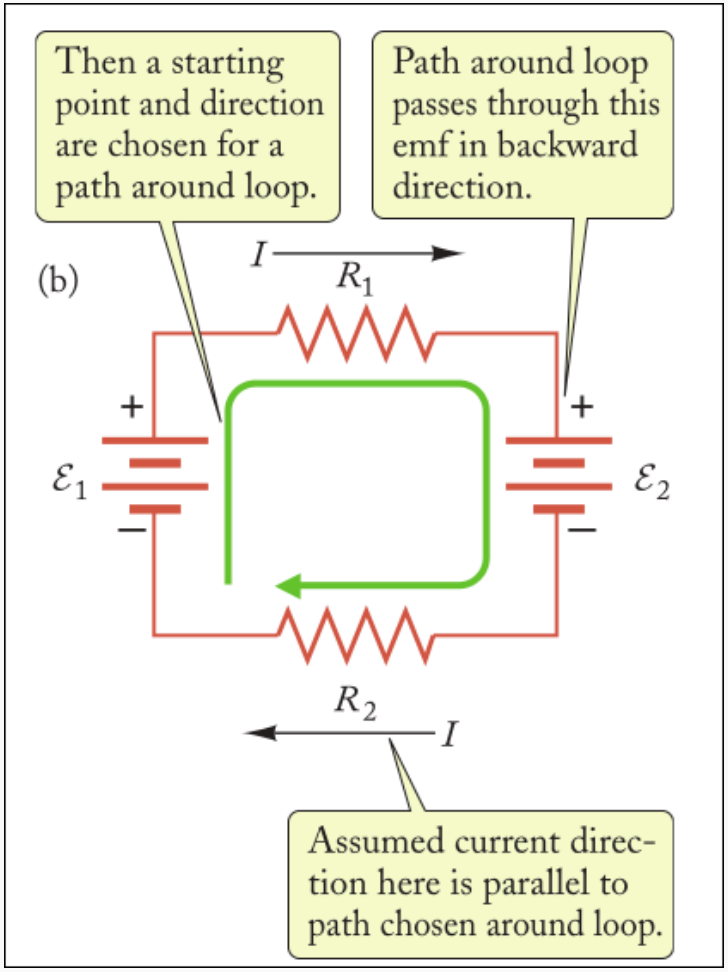
$$\varepsilon_1 - IR_1 - \varepsilon_2 - IR_2 = 0$$

$$I = \frac{\varepsilon_1 - \varepsilon_2}{R_1 + R_2}$$

For $\varepsilon_1 = 12\ \text{V}$, $\varepsilon_2 = 15\ \text{V}$, $R_1 = 4\ \Omega$, and $R_2 = 2\ \Omega$,

$$I = -0.5\ \text{A}$$

The current actually flows counterclockwise.



Multiloop circuits

Multiloop circuits: loop method

To solve these circuits, we need a simultaneous set of equations with the currents as unknowns.

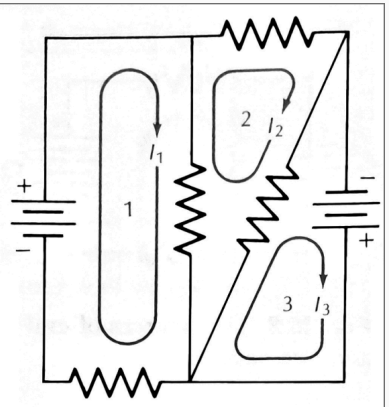


Fig. 29.10 A multiloop circuit. The currents I_1, I_2, and I_3 are regarded as flowing in closed loops. Note that loops 1 and 2 share the middle resistor; and loops 2 and 3 share the diagonal resistor. The net current in the middle resistor is $I_1 - I_2$, and the net current in the diagonal resistor is $I_2 - I_3$.

Loop method

(a) Label the loop currents $I_1, I_2, \ldots$ and arbitrarily assign directions.

(b) Regard the circuit as a collection of closed current loops. The loops may overlap, but each must have a section that does not overlap.

(c) Apply Kirchhoff's voltage rule to each loop. Note that if a resistor belong to two adjacent loops, then net current is sum of the two loop currents.

Shared resistors

The sum of all emfs and potential drops must be zero around each loop.

If a resistor belongs to two adjacent loops, its net current is the algebraic sum of the two loop currents.

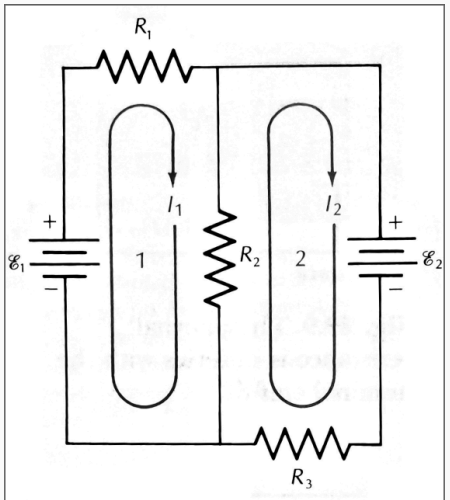


Fig. 29.11 The loops 1 and 2 share the resistor R_2. The currents I_1 and I_2 flow through R_2 in opposite directions; the net current through R_2 is therefore the difference between I_1 and I_2.

For the example shown,

$$\varepsilon_1 - I_1 R_1 - I_1 R_2 + I_2 R_2 = 0$$

$$-\varepsilon_2 - I_2 R_3 - I_2 R_2 + I_1 R_2 = 0$$

Example: solving the loop equations

For $\varepsilon_1 = 12\ \text{V}$, $\varepsilon_2 = 8\ \text{V}$, $R_1 = 4\ \Omega$, $R_2 = 4\ \Omega$, and $R_3 = 2\ \Omega$,

$$12 - 8I_1 + 4I_2 = 0$$

$$-8 - 6I_2 + 4I_1 = 0$$

Solving simultaneously,

$$I_1 = 1.25\ \text{A}, \quad I_2 = -0.5\ \text{A}$$

Kirchhoff's junction rule

There is an alternative way using Kirchhoff's first rule:

> The sum of all currents entering any branch point of the circuit must equal the sum of the currents leaving.

This rule expresses conservation of charge.

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

Multiloop circuits: branch method

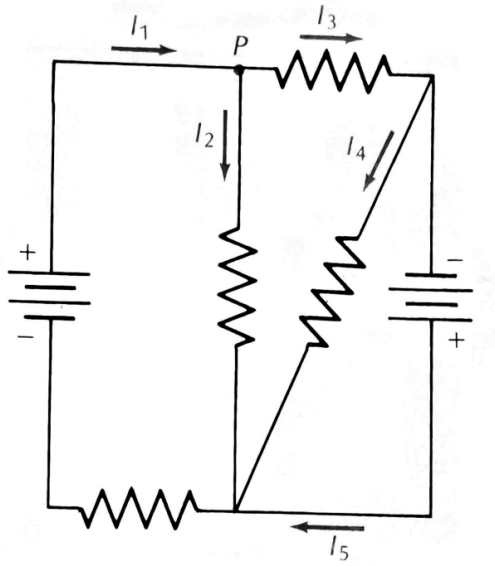


Fig. 29.12 A multiloop circuit. The currents in the distinct branches of the circuit are I_1, I_2, I_3, I_4, and I_5. The point P is a branch point, where three wires merge.

Branch method

(a) Regard the circuit as a collection of branches beginning and ending at points where wires merge.

(b) Label the distinct branch currents $I_1, I_2, I_3, \ldots$ and arbitrarily assign directions.

(c) Apply the loop rule: the sum of emfs and potential drops is zero.

(d) Apply the junction rule at branch points.

Example: branch equations

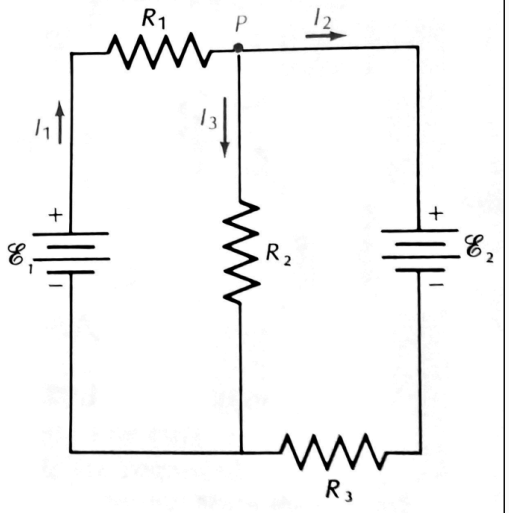


Fig. 29.13 The currents in the distinct branches of this circuit are I_1, I_2, and I_3.

Loop 1:

$$\varepsilon_1 - I_1 R_1 - I_3 R_2 = 0$$

Loop 2:

$$-\varepsilon_2 - I_2 R_3 + I_3 R_2 = 0$$

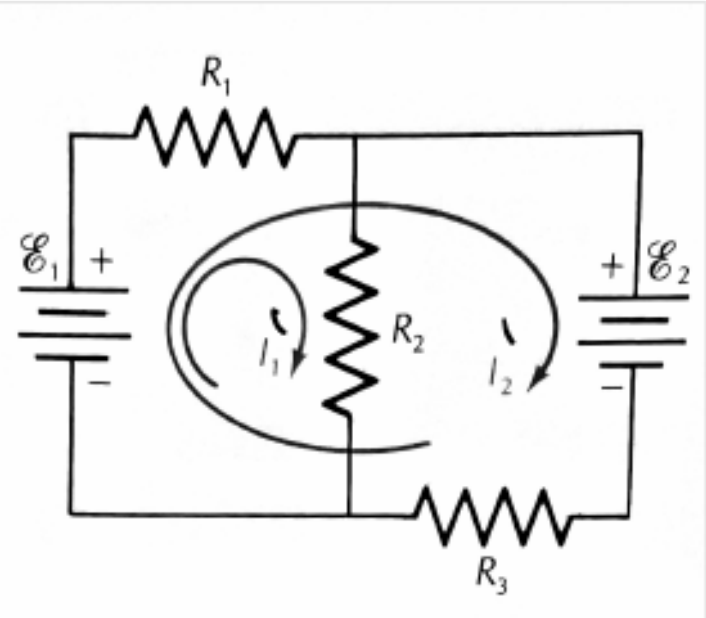
At the junction:

$$I_1 = I_3 + I_2$$

The loop method involves fewer equations and is usually quicker. The branch method gives the actual currents directly.

An alternative choice of loops

We can even use the following loops:



$$\varepsilon_1 - (I_1' + I_2')R_1 - I_1'R_2 = 0$$

$$\varepsilon_1 - (I_1' + I_2')R_1 - I_2'R_3 - \varepsilon_2 = 0$$

Comparing with the first solution,

$$I_1' + I_2' = I_1, \quad I_2' = I_2$$

Energy in circuits

Energy in circuits

If a charge $\mathrm{d}q$ passes through a source of emf ε, the work done is

$$\mathrm{d}W = \varepsilon\,\mathrm{d}q$$

$$\frac{\mathrm{d}W}{\mathrm{d}t} = \varepsilon\frac{\mathrm{d}q}{\mathrm{d}t}$$

$$P = \varepsilon I$$

In the previous section, we found that the power dissipated is $P = \Delta V I$. For an ohmic device,

$$P = I^2 R = \frac{(\Delta V)^2}{R}$$

RC circuits

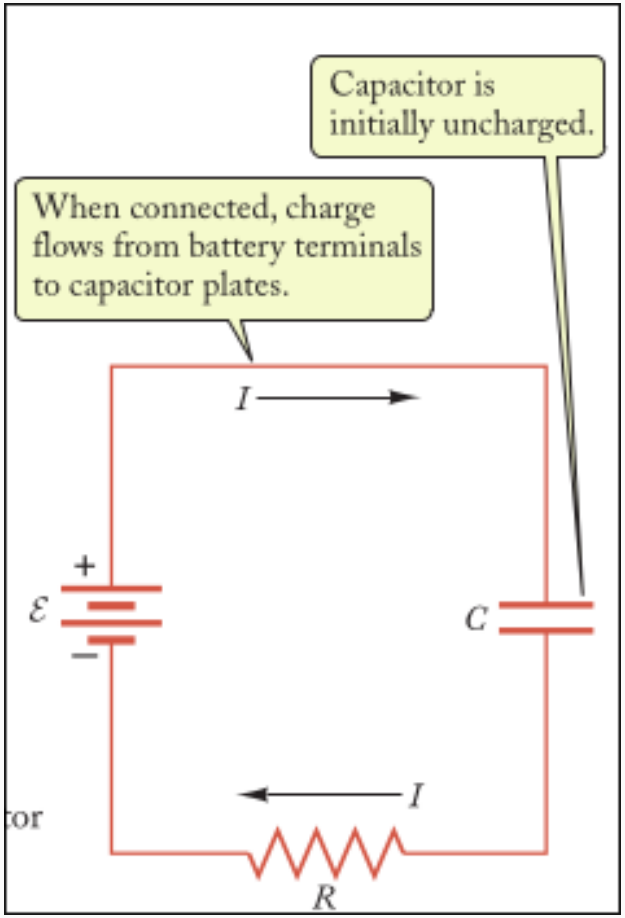
RC circuits

Throughout this chapter, we have dealt with steady, time-independent currents. Kirchhoff's rules also apply to time-dependent currents, as long as the emfs and currents do not vary quickly.

A rough criterion is that currents and emfs must not change significantly during a time interval equal to the travel time of a light signal around the circuit.

Charging a capacitor

Assume the capacitor is uncharged at $t = 0$: $q(0) = 0$.



$$\varepsilon - IR - \frac{q}{C} = 0$$

Since $I = \frac{dq}{dt}$,

$$\varepsilon - R\frac{dq}{dt} - \frac{q}{C} = 0$$

$$\frac{dq}{dt} = \frac{\varepsilon C - q}{RC}$$

Charging: integrating the equation

Separate the variables:

$$\frac{\mathrm{d}q}{\varepsilon C - q} = \frac{\mathrm{d}t}{RC}$$

$$\int_0^q \frac{\mathrm{d}q'}{\varepsilon C - q'} = \frac{1}{RC}\int_0^t \mathrm{d}t'$$

$$-\ln(C\varepsilon - q) + \ln(C\varepsilon) = \frac{t}{RC}$$

$$\ln\left(1 - \frac{q}{C\varepsilon}\right) = -\frac{t}{RC}$$

$$q(t) = C\varepsilon\left(1 - e^{-\frac{t}{RC}}\right), \quad I = \frac{\mathrm{d}q}{\mathrm{d}t} = \frac{\varepsilon}{R}e^{-\frac{t}{RC}}$$

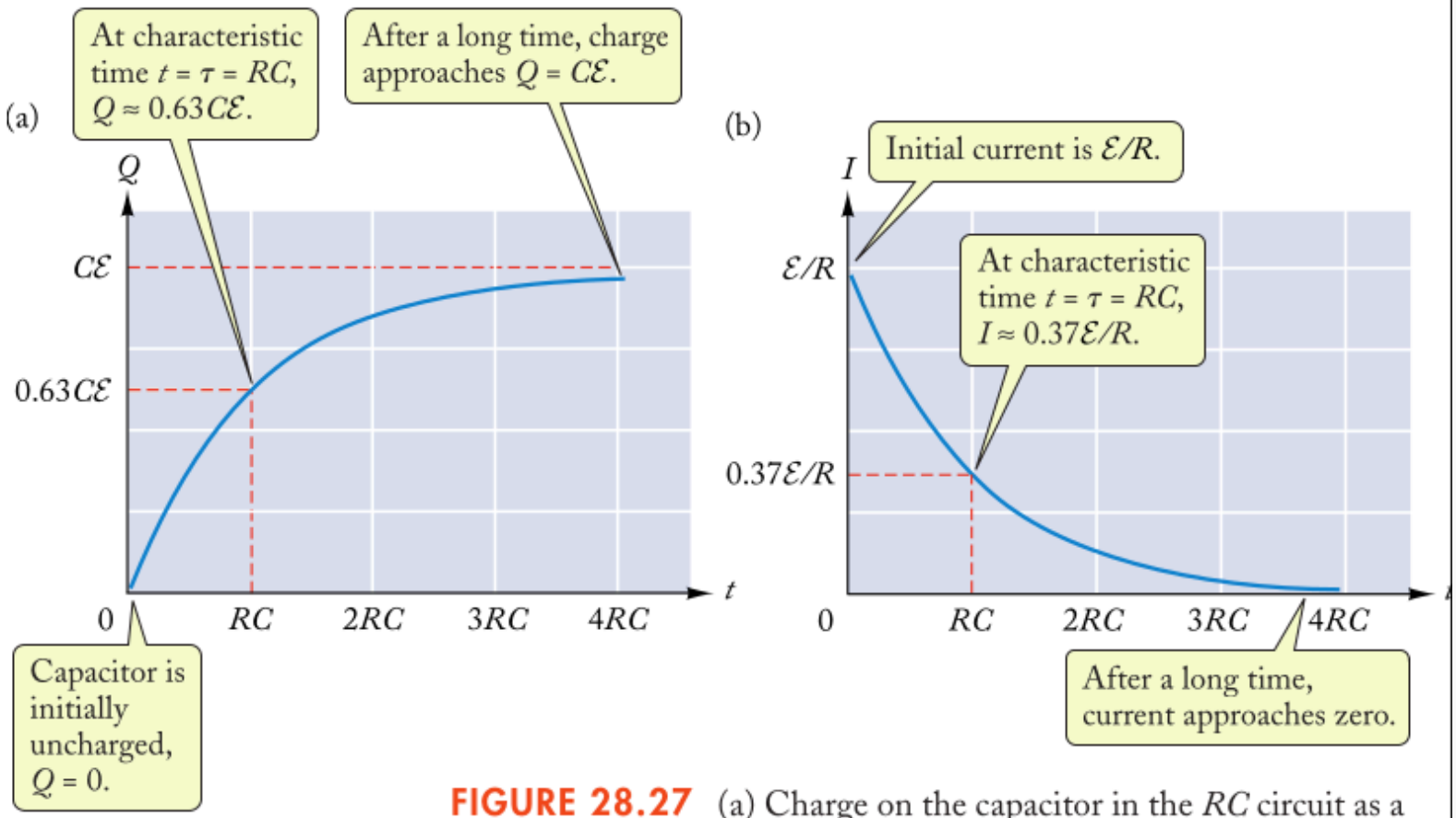
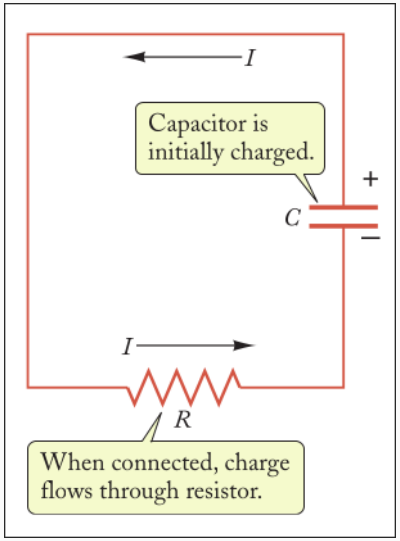


FIGURE 28.27 (a) Charge on the capacitor in the *RC* circuit as a function of time. (b) Current in the *RC* circuit as a function of time.

Discharging a capacitor

After charging is complete, disconnect the source and connect the capacitor and resistor together.



$$-IR - \frac{q}{C} = 0$$

$$-R\frac{dq}{dt} - \frac{q}{C} = 0$$

$$q(t) = \varepsilon C e^{-\frac{t}{RC}}$$

$$I = \frac{dq}{dt} = -\frac{\varepsilon}{R} e^{-\frac{t}{RC}}$$

The current is negative because it is opposite to the original current.

Discharging: voltage and current

$$V(t) = \varepsilon e^{-\frac{t}{RC}}, \quad I(t) = -\frac{\varepsilon}{R} e^{-\frac{t}{RC}}$$

